

Adapting Subtitles: Towards Accessible Subtitled Media for People with Aphasia

Zihao You

University of Dundee
Dundee, Scotland, United Kingdom
2565278@dundee.ac.uk

Abstract

Subtitles can be consumed via TVs, laptops, and smartphones, which play a key role in many people's daily lives. However, this marginalises people with complex accessibility needs in the context of comprehension, such as people with aphasia, a communication and language impairment. To address this accessibility challenge with a human-centred design approach, I started by building rapport with the community of focus and involving them in all stages of the design process. This paper reports on ongoing doctoral work that explores how personalising and customising subtitled media can fulfil complex accessibility needs for communities with aphasia, by delineating the research background and motivation, the accomplished work, and the proposed next steps.

CCS Concepts

• Human-centered computing → Accessibility technologies.

Keywords

Subtitles; Closed Captions; Adaptive Media; Aphasia; Participatory Design

ACM Reference Format:

Zihao You. 2025. Adapting Subtitles: Towards Accessible Subtitled Media for People with Aphasia. In *Companion of the Computer-Supported Cooperative Work and Social Computing (CSCW Companion '25)*, October 18–22, 2025, Bergen, Norway. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3715070.3747337>

1 Introduction

Subtitles/Closed Captions¹ correspond to texts that illustrate what is being spoken on a film or television [51]. Subtitled media consumption has recently become increasingly prominent due to the proliferation of digital devices with more diverse contexts of use and modalities. However, the current one-size-fits-all approach to subtitle display creates accessibility challenges. To date, a majority of accessible subtitle interventions are tailored to blind and visually impaired (BVI) people and deaf and hard of hearing (DHH) people only [37], leaving other communities little explored.

¹Closed captions (CC) also provide a text description of sound effects. Most streaming sites only have the option for 'English[CC]' for English subtitles and therefore in this work, we collectively refer to both as subtitles.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CSCW Companion '25, Bergen, Norway

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1480-1/25/10

<https://doi.org/10.1145/3715070.3747337>

Aphasia is a disorder of language comprehension and formulation (i.e., understanding, speaking, reading, writing and using numbers [47]) caused by malfunction in certain brain areas [19], which affects approximately one-third of stroke patients [2, 47], people with severe head injury or brain tumours [38]. There are currently at least 15 million people worldwide who have aphasia [21]. While there has been research focusing on accessibility interventions for people with aphasia (PWA), this research has concentrated mostly on PWA with speech production impairments. Yet, interventions addressing PWA with reading impairments remain scarce and underrepresented [13].

This doctoral work, therefore, seeks to close accessibility gaps between PWA with comprehension deficits and subtitled media.

2 Background and related work

2.1 Consumption and adaptation of subtitled media

The design of traditional subtitles, as suggested by industry guidelines, is normally white text surrounded by a black background at the bottom centre of the screen [4], which offers people alternatives to access the audio material inside media. People with accessibility needs, however, face additional challenges during consumption. This has been highlighted by an array of recent works that revolve around BVI people [37] and DHH people [5, 37, 43, 44, 57] as the target community. Potential consequences of depression and exclusion from society [57] have ignited researchers' ambitions of innovating subtitle displays to balance information conveyance and viewer engagement.

Over the past three decades, research has consistently sought to refine subtitle presentation to more effectively address diverse user requirements, including accessibility challenges encountered by disabled populations. Brett [8] in 1995 used hypertext that triggers on-click explanations for sophisticated vocabulary within subtitles to enhance people's listening skills in Business English. Subtitles were altered using animation by Rashid et al. [42] in the mid-2000s, which captured characters' emotions. The adaptations began to diversify during the 2010s, with several works attempting to locate subtitles outside the traditional viewing window in virtual reality. Four subtitle behaviours were proposed by BBC Research & Development [9], in which users preferred subtitles moving with their head position [10] and with 12.5 degrees below the eye-line as an optimum angle [45]. Furthermore, Kurzahls et al. [22] presented two gaze-adaptive methods that leveraged users' gaze to determine the subtitle placement. Wearable vibrotactile interaction [56] was presented, which translates speakers' voices into vibrations, enabling people with DHH to feel the vibrations on their wrists and engage with the content.

2.2 Aphasia and state-of-the-art accessibility interventions

Augmentative and alternative communication (AAC) strategies are typical examples that supplement ordinary communication methods that are impaired [28, 31]. Several notable works exist, including the hybrid desktop-handheld system that enables PWA to develop speech communication using graphics and sound on a desktop computer [7]. Wearable AAC devices have also grown in prominence. Three high-fidelity prototypes [18] and two low-input prototypes [15] designed by Humphrey et al. supported various body placements, helping PWA navigate daily life.

Non-/indirect AAC interventions for PWA should also be emphasised. The INCA project [11] proposed novel interaction techniques that facilitated PWA's creation and curation of digital content. For instance, by picking a source text, users can perform parameterised redaction to rearrange words, create new sentences, and share with others [34]. The creation of comic strips has been supported by pre-customising the comic's length and themes [49]. Moreover, expressive multimedia content creation can be achieved by selecting and rearranging tangible objects on *CreaTable*, which can be 'played' audiovisually on the screen [35]. Collaborations with artists with aphasia also made creating digital art possible by taking advantage of previously created physical material [33].

Although an increasing trend has been found for accessibility research in subtitled media in the last few decades [27, 37], none of them has included PWA as the community of focus. While progress has been made in several areas, including designing accessibility interventions for PWA in direct communication and digital content creation, a substantial opportunity persists in addressing PWA's needs in subtitle consumption. The growing significance of digital inclusion and the complexity of subtitle presentation make it crucial to understand the challenges of adapting media to meet their requirements, which is where my overall research direction pinpoints.

3 Research Questions

The primary goal of this PhD work would be to explore how subtitled media can be adapted to enhance the PWA's experience of media consumption. As such, I address the following three research questions:

RQ1 *What factors influence subtitle location, position, content, and styling to enhance media accessibility?*

This question has been inspired by looking past the history of adapting subtitled media in terms of location, position, content, and styling to investigate the factors involved in creating adaptations thoroughly.

RQ2 *What challenges do PWA face when consuming subtitled media and in what ways can adaptation be used to better support their consumption of subtitled media?*

This question examines the accessibility and usability challenges PWA encounter in subtitled media consumption, and explores preferences regarding the potential applicability of subtitle adaptation intervention.

RQ3 *How can PWA's experience of consuming adapted subtitled media be evaluated?*

Grounded in the findings derived from **RQ2**, this question

will assess PWA's interactions with the developed subtitle adaptation intervention.

4 Methodology and Contributions to date

RQ1 is addressed by the outcomes of a systematic literature review (SLR) that understand the past challenges faced by various communities of focus when consuming subtitled media. To tackle **RQ2**, the research narrows down to PWA as the target end-user from the research gaps identified during SLR, leveraging a combination of **Design Thinking** [41] and **Co-design** [46] approaches. This strategy involves users throughout the entire project cycle, which is gaining traction in accessibility research focused on addressing challenges faced by older people [26] and disabled communities [16, 25, 36]. It allows for the direct engagement of PWA via surveys and interviews [58] to develop accessibility interventions that better meet their requirements. The final phase of this doctoral work is to evaluate the experience of consuming the adapted subtitle media and examine the system's usability and potential for long-term adoption, which answers **RQ3**. Four main contributions will be made in total, with categories derived from Wobbrock and Kientz [59].

4.1 Contribution I - SLR of access to subtitled media (Survey)

There was no survey of the accessibility research in subtitled media. Previous SLRs have examined subtitles for language acquisition [5, 40, 43] or subtitling tasks comparison [29]. Others have focused on accessibility research and media consumption in a broader scope. For instance, Mack et al. [27] conducted a survey of accessibility papers in CHI and ASSETS, Vatavu [53] considered television and media accessibility papers that appeared in IMX/TVX, Nevesky et al. [37] researched digital audiovisual media accessibility. SLRs can also be tailored to specific communities, such as those exploring the improvements of object-based audio accessibility for hearing-impaired users [57].

This SLR aims to investigate the obstacles and opportunities of adapting subtitled media for people with various accessibility needs, following guidelines from Boland et al. [6]. Early scoping searches have been performed and used as a gauge for constructing **RQ1**. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) procedure [24, 30] has then been followed, which facilitates the narrowing of the literature base used in the final analysis. The dataset includes 147 papers covering the 29-year period (1995-2024). Inspired by previous SLRs of accessibility research [17, 27, 37], a codebook has been created to capture data (via codification) from papers qualitatively. This answered **RQ1** on the aspects of broadcasting method, viewing device, targeting community and user study location. To elaborate, at least two-thirds ($N = 102$) of publications had subtitles broadcast as videos, while few were broadcast as live streams ($N = 4$). 85% ($N = 125$) of papers had subtitles displayed on desktops/laptops, whereas less than 5% ($N = 6$) of papers had subtitles displayed on big screens. The two most explored communities of focus refer to DHH ($N = 58$) and BVI ($N = 19$), but no work has been found exploring PWA. Researchers preferred involving users in lab settings rather than in their residential homes. Additionally, a taxonomy for the types

of subtitle adaptation explored in various communities has been created. The results from this SLR will be further reported in an upcoming journal article.

5 Planned future work

Based upon the potential research gaps identified from the SLR, it is possible to start addressing **RQ2** and **RQ3**. The Double Diamond design thinking process [20] will be followed to add rigour to the overall design strategy.

5.1 Contribution II - Requirements Elicitation from SLTs & PWA (*Empirical & Theoretical*)

A qualitative approach is proposed to understand the social, physical, and psychological factors that influence PWA's consumption of subtitled media. Conducting exploratory interviews with speech and language therapists (SLTs) [39] enables a thorough investigation of **RQ2**. Given PWA's impairments in terms of text comprehension, a follow-up survey [3] consisting of closed questions will be adopted. This offers potential insights regarding the challenges they face when consuming subtitles. All participants will be recruited from local charities, and ethical approval [54] will be secured before gathering human data. The findings synthesised from exploratory interviews and surveys contribute to the design guidance for subtitle adaptation interventions tailored to PWA.

5.2 Contribution III - Design and Development of Accessibility Intervention (*Artifact*)

Grounded in design guidelines derived from the interview and the survey, the solution's main features will evolve, which is where the implementation will begin. The creation of personas (modelling how different groups of users may use our system through a visual prop) [32] will be considered to capture their needs better. The main challenge of such interventions is whether the user is currently focusing on the subtitled text. Eye-tracking devices (e.g., Tobii²) can be utilised to monitor their eye movements while consuming subtitled media [50]. The generated heatmaps [55], as a result, reveal when the subtitle adaptations will be triggered. A control group [12, 23] of non-disabled people will be included, which can be used as baseline eye gaze data to differentiate the eye movements. Besides, seeking input from professional subtitlers during design iterations may improve the system's robustness.

5.3 Contribution IV - Testing and Evaluation of Accessibility Intervention (*Empirical*)

As horizontal prototyping [1] will be carried out throughout the implementation of the accessibility interventions, the testing and evaluation will only begin once most of the system features are functional. To minimise the likelihood of confounding factors [48] that may impact the research finding, a mixed-method approach will be applied. Comparing our system with other existing subtitle accessibility interventions (from the SLR database) pairwise may help better understand the end user's change of experience. They can use an evaluation sheet [52] to decide their preference by selecting one of the two subtitle displays or neutral. This will

be followed by semi-structured interviews [14], enabling them to further reason about their responses.

References

- [1] Philip Bacon, Reinhard Budde, Karlheinz Kautz, Karin Kuhlenkamp, and Heinz Züllighoven. 2012. *Prototyping: An Approach to Evolutionary System Development* (1st. ed.). Springer-Verlag, Berlin, Heidelberg.
- [2] Morgan Bailey, Lauren Burningham, and Erin Fifield. 2018. *The Efficacy of Tele-practice on Expressive Language Outcomes for Adults with Aphasia: A Systematic Review*. University of Vermont. Retrieved April 03, 2025 from file:///Users/andersonyou/Downloads/The%20Efficacy%20of%20Tele-practice%20on%20Expressive%20Language%20Outcomes%20for%20(3).pdf
- [3] Mai Baunstrup and Lars Bo Larsen. 2013. Elderly's barriers and requirements for interactive TV. In *Universal Access in Human-Computer Interaction. User and Context Diversity: 7th International Conference, UAHCI 2013, Held as Part of HCI International 2013, Las Vegas, NV, USA, July 21-26, 2013, Proceedings, Part II 7 (UAHCI '13)*. Springer, Berlin/Heidelberg, Germany, 13–22. doi:10.1007/978-3-642-39191-0_2
- [4] BBC. 2024. *Subtitle Guidelines*. BBC. Retrieved April 03, 2025 from https://www.bbc.co.uk/accessibility/forproducts/guides/subtitles/
- [5] Sharon Black. 2021. The potential benefits of subtitles for enhancing language acquisition and literacy in children: An integrative review of experimental research. *Translation, Cognition & Behavior* 4, 1 (Aug. 2021), 82. doi:10.1075/tcb.00051.bla
- [6] Angela Boland, Gemma Cherry, and Rumona Dickson. 2017. *Doing a systematic review: A student's guide* (2nd. ed.). SAGE Publications Ltd, Thousand Oaks, California.
- [7] Jordan L. Boyd-Graber, Sonya S. Nikolova, Karyn A. Moffatt, Kenrick C. Kin, Joshua Y. Lee, Lester W. Mackey, Marilyn M. Tremaine, and Maria M. Klawe. 2006. Participatory design with proxies: developing a desktop-PDA system to support people with aphasia. In *Proceedings of the SIGCHI conference on Human Factors in computing systems (CHI '06)*. Association for Computing Machinery, New York, NY, USA, 151. doi:10.1145/1124772.1124797
- [8] Paul Brett. 1995. Multimedia for listening comprehension: The design of a multimedia-based resource for developing listening skills. *System* 23, 1 (Feb. 1995), 77–85. doi:10.1016/0346-251X(94)00054-A
- [9] Andy Brown, Jayson Turner, Jake Patterson, Anastasia Schmitz, Mike Armstrong, and Maxine Glancy. 2017. Subtitles in 360-degree Video. In *Adjunct Publication of the 2017 ACM International Conference on Interactive Experiences for TV and Online Video (TVX '17)*. Association for Computing Machinery, New York, NY, USA, 5–6. doi:10.1145/3084289.3089915
- [10] Andy Brown, Jayson Turner, Jake Patterson, Anastasia Schmitz, Mike Armstrong, and Maxine Glancy. 2018. *Exploring subtitle behaviour for 360 video*. BBC. Retrieved April 03, 2025 from https://downloads.bbc.co.uk/rd/pubs/whp/whp-pdf-files/WHP330.pdf
- [11] City University of London Centre for HCI Design. 2021. *Inclusive Digital Content for People with Aphasia*. City University of London. Retrieved April 03, 2025 from https://blogs.city.ac.uk/inca/
- [12] Agnes S Chan, Tsz-Lok Lee, Sophia L Sze, Natalie S Yang, and Yvonne MY Han. 2022. Eye-tracking training improves the learning and memory of children with learning difficulty. *Scientific Reports* 12, 1 (Aug. 2022), 13974. doi:10.1038/s41598-022-18286-6
- [13] Giorgia Cistola, Mireia Farrús, and Ineke van der Meulen. 2021. Aphasia and acquired reading impairments: What are the high-tech alternatives to compensate for reading deficits? *International Journal of Language & Communication Disorders* 56, 1 (Jan. 2021), 161–162. doi:10.1111/1460-6984.12569
- [14] Michael Crabb, Rhianne Jones, Mike Armstrong, and Chris J Hughes. 2015. Online news videos: the UX of subtitle position. In *Proceedings of the 17th international ACM SIGACCESS conference on Computers & accessibility (ASSETS '15)*. Association for Computing Machinery, New York, NY, USA, 218. doi:10.1145/2700648.2809866
- [15] Humphrey Curtis, Ying Hei Lau, and Timothy Neate. 2024. Breaking Badge: Augmenting Communication with Wearable AAC Smartbadges and Displays. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, 2. doi:10.1145/3613904.3642327
- [16] Humphrey Curtis and Timothy Neate. 2023. Watch Your Language: Using Smartwatches To Support Communication. In *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '23)*. Association for Computing Machinery, New York, NY, USA, 1–21. doi:10.1145/3597638.3608379
- [17] Humphrey Curtis, Timothy Neate, and Carlota Vazquez Gonzalez. 2022. State of the art in AAC: A systematic review and taxonomy. In *Proceedings of the 24th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '22)*. Association for Computing Machinery, New York, NY, USA, 1–22. doi:10.1145/3517428.3544810

²<https://www.tobii.com/products/eye-trackers>

- [18] Humphrey Curtis, Zihao You, William Deary, Miruna-Ioana Tudoreanu, and Timothy Neate. 2023. Envisioning the (in) visibility of discreet and wearable AAC devices. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA, 2. doi:10.1145/3544548.3580936
- [19] Antonio R. Damasio. 1992. Aphasia. *New England Journal of Medicine* 326, 8 (Feb. 1992), 531. doi:10.1056/NEJM199202203260806
- [20] Jeff Humble. 2024. *What is the Double Diamond Design Process?* The Fountain Institute. Retrieved April 03, 2025 from <https://www.thefountaininstitute.com/blog/what-is-the-double-diamond-design-process>
- [21] Maria V. Ivanova and Nina F. Dronkers. 2022. Aphasia: How our language system can “break”. *Frontiers for young minds* 10 (April 2022), 626477. doi:10.3389/frym.2022.626477
- [22] Kuno Kurzhals, Fabian Göbel, Katrin Angerbauer, Michael Sedlmair, and Martin Raubal. 2020. A view on the viewer: Gaze-adaptive captions for videos. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. Association for Computing Machinery, New York, NY, USA, 1–12. doi:10.1145/3313831.3376266
- [23] Tsz Lok Lee, Michael K Yeung, Sophia L Sze, and Agnes S Chan. 2021. Eye-tracking training improves inhibitory control in children with attention-deficit/hyperactivity disorder. *Brain sciences* 11, 3 (March 2021), 314. doi:10.3390/brainsci11030314
- [24] Alessandro Liberati, Douglas G Altman, Jennifer Tetzlaff, Cynthia Mulrow, Peter C Gøtzsche, John PA Ioannidis, Mike Clarke, Philip J Devereaux, Jos Kleijnen, and David Moher. 2009. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: explanation and elaboration. *Annals of internal medicine* 151, 4 (Aug. 2009), W-65–W-94. doi:10.7326/0003-4819-151-4-200908180-00136
- [25] Stephen Lindsay, Katie Brittain, Daniel Jackson, Cassim Ladha, Karim Ladha, and Patrick Olivier. 2012. Empathy, participatory design and people with dementia. In *Proceedings of the SIGCHI conference on Human factors in computing systems (CHI '12)*. Association for Computing Machinery, New York, NY, USA, 521–530. doi:10.1145/2207676.2207749
- [26] Stephen Lindsay, Daniel Jackson, Guy Schofield, and Patrick Olivier. 2012. Engaging older people using participatory design. In *Proceedings of the SIGCHI conference on human factors in computing systems (CHI '12)*. Association for Computing Machinery, New York, NY, USA, 1199–1208. doi:10.1145/2207676.2208570
- [27] Kelly Mack, Emma McDonnell, Dhruv Jain, Lucy Lu Wang, Jon E. Froehlich, and Leah Findlater. 2021. What do we mean by “accessibility research”? A literature survey of accessibility papers in CHI and ASSETS from 1994 to 2019. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. Association for Computing Machinery, New York, NY, USA, 1. doi:10.1145/3411764.3445412
- [28] Sally Millar and Janet Scott. 1998. *What is augmentative and alternative communication? An introduction*. University of Edinburgh. Retrieved April 03, 2025 from <https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=52f220ecc9ac08b8d6d20a82f73fb4efa2c8aa09>
- [29] Daniela Mocetuma, Tania Ramirez-delReal, Guillermo Ruiz, and Othón González-Chávez. 2023. Video captioning: a comparative review of where we are and which could be the route. *Computer Vision and Image Understanding* 231 (June 2023), 103671. doi:10.1016/j.cviu.2023.103671
- [30] David Moher, Alessandro Liberati, Jennifer Tetzlaff, Douglas G Altman, and the PRISMA Group*. 2009. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *Annals of internal medicine* 151, 4 (Aug. 2009), 264–269. doi:10.7326/0003-4819-151-4-200908180-00135
- [31] A. Moorcroft, N. Scarinci, and C. Meyer. 2019. A systematic review of the barriers and facilitators to the provision and use of low-tech and unaided AAC systems for people with complex communication needs and their families. *Disability and Rehabilitation: Assistive Technology* 14, 7 (Oct. 2019), 710–731. doi:10.1080/17483107.2018.1499135
- [32] Timothy Neate, Aikaterini Bourazeri, Abi Roper, Simone Stumpf, and Stephanie Wilson. 2019. Co-created personas: Engaging and empowering users with diverse needs within the design process. In *Proceedings of the 2019 CHI conference on human factors in computing systems (CHI '19)*. Association for Computing Machinery, New York, NY, USA, 1–12. doi:10.1145/3290605.3300880
- [33] Timothy Neate, Abi Roper, and Stephanie Wilson. 2020. Painting a Picture of Accessible Digital Art. In *Proceedings of the 22nd International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '20)*. Association for Computing Machinery, New York, NY, USA, 1–3. doi:10.1145/3373625.3418019
- [34] Timothy Neate, Abi Roper, Stephanie Wilson, and Jane Marshall. 2019. Empowering expression for users with aphasia through constrained creativity. In *Proceedings of the 2019 CHI conference on human factors in computing systems (CHI '19)*. Association for Computing Machinery, New York, NY, USA, 6. doi:10.1145/3290605.3300615
- [35] Timothy Neate, Abi Roper, Stephanie Wilson, Jane Marshall, and Madeline Cruice. 2020. CreaTable content and tangible interaction in Aphasia. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. Association for Computing Machinery, New York, NY, USA, 3–4. doi:10.1145/3313831.3376490
- [36] Alexandre Nevsky, Timothy Neate, Elena Simperl, and Madeline N Cruice. 2024. Lights, Camera, Access: A Closeup on Audiovisual Media Accessibility and Aphasia. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, 1–17. doi:10.1145/3613904.3641893
- [37] Alexandre Nevsky, Timothy Neate, Elena Simperl, and Radu-Daniel Vatavu. 2023. Accessibility Research in Digital Audiovisual Media: What Has Been Achieved and What Should Be Done Next?. In *Proceedings of the 2023 ACM International Conference on Interactive Media Experiences (IMX '23)*. Association for Computing Machinery, New York, NY, USA, 94–114. doi:10.1145/3573381.3596159
- [38] NHS. 2025. *Aphasia*. NHS. Retrieved May 12, 2025 from <https://www.nhs.uk/conditions/aphasia/>
- [39] NHS. 2025. *Speech and language therapist*. NHS. Retrieved April 03, 2025 from <https://www.healthcareers.nhs.uk/explore-roles/allied-health-professionals/roles-allied-health-professions/speech-and-language-therapist>
- [40] Maribel Montero Perez, Wim Van Den Noortgate, and Piet Desmet. 2013. Captioned video for L2 listening and vocabulary learning: A meta-analysis. *System* 41, 3 (Aug. 2013), 720. doi:10.1016/j.system.2013.07.013
- [41] Hasso Plattner. 2025. *An Introduction to Design Thinking PROCESS GUIDE*. Institute of Design at Stanford. Retrieved April 03, 2025 from <https://web.stanford.edu/~mshanks/MichaelShanks/files/509554.pdf>
- [42] Raisa Rashid, Jonathan Aitken, and Deborah I. Fels. 2006. Expressing emotions using animated text captions. In *Computers Helping People with Special Needs: 10th International Conference, ICCHP 2006, Linz, Austria, July 11–13, 2006. Proceedings 10 (ICCHP '06)*. Springer, Berlin, Heidelberg, 24–31. doi:10.1007/11788713_5
- [43] Barry Lee Reynolds, Yanwen Cui, Chian-Wen Kao, and Nathan Thomas. 2022. Vocabulary acquisition through viewing captioned and subtitled video: A scoping review and meta-analysis. *Systems* 10, 5 (Aug. 2022), 134. doi:10.3390/systems10050133
- [44] RNID. 2001. *Research into the demand for Welsh language subtitling in Wales*. S4C. Retrieved April 03, 2025 from https://www.s4c.cymru/abouts4c/corporate/pdf/e_adroddiad_isdeitlo.pdf
- [45] Sylvia Rothe, Kim Tran, and Heinrich Hussmann. 2018. Positioning of Subtitles in Cinematic Virtual Reality. In *International Conference on Artificial Reality and Telexistence and Eurographics Symposium on Virtual Environments (ICAT-EGVE '18)*. The Eurographics Association, Eindhoven, Netherlands, 3. doi:10.2312/egve.20181307
- [46] Elizabeth B.-N. Sanders and Pieter Jan Stappers. 2008. Co-creation and the new landscapes of design. *Co-design* 4, 1 (March 2008), 5–18. doi:10.1080/15710880701875068
- [47] Sue Sherratt. 2024. People with aphasia living alone: A scoping review. *Aphasiology* 38, 4 (April 2024), 713. doi:10.1080/02687038.2023.2227403
- [48] Andrea C Skelly, Joseph R Dettori, and Erika D Brodt. 2012. Assessing bias: the importance of considering confounding. *Evidence-based spine-care journal* 3, 01 (Feb. 2012), 9–12. doi:10.1055/s-0031-1298595
- [49] Carla Tamburro, Timothy Neate, Abi Roper, and Stephanie Wilson. 2020. Accessible creativity with a comic spin. In *Proceedings of the 22nd International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '20)*. Association for Computing Machinery, New York, NY, USA, 3–5. doi:10.1145/3373625.3417012
- [50] Enkelelda Thaqi, Mohamed Omar Mantawy, and Enkelelda Kasneci. 2024. SARA: Smart AI Reading Assistant for Reading Comprehension. In *Proceedings of the 2024 Symposium on Eye Tracking Research and Applications (ETRA '24)*. Association for Computing Machinery, New York, NY, USA, 1–3. doi:10.1145/3649902.3655661
- [51] Cambridge Advanced Learner's Dictionary & Thesaurus. 2025. *Meaning of subtitles in English*. Cambridge University Press & Assessment. Retrieved April 03, 2025 from <https://dictionary.cambridge.org/dictionary/english/subtitles>
- [52] Máté Ákos Tündik, Balázs Tarján, and György Szaszák. 2020. A low latency sequential model and its user-focused evaluation for automatic punctuation of ASR closed captions. *Computer Speech & Language* 63 (Sept. 2020), 15–17. doi:10.1016/j.csl.2020.101076
- [53] Radu-Daniel Vatavu. 2021. Accessibility of interactive television and media experiences: users with disabilities have been little voiced at IMX and TVX. In *Proceedings of the 2021 ACM International Conference on Interactive Media Experiences (IMX '21)*. Association for Computing Machinery, New York, NY, USA, 218–222. doi:10.1145/3452918.3465485
- [54] Susan Walker and Sue Read. 2011. Accessing vulnerable research populations: an experience with gatekeepers of ethical approval. *International journal of palliative nursing* 17, 1 (Jan. 2011), 14–18. doi:10.12968/ijpn.2011.17.1.14
- [55] Jun Wang, Sichen Li, Ke Xue, and Li Chen. 2022. What is the competence boundary of Algorithms? An institutional perspective on AI-based video generation. *Displays* 73 (July 2022), 102240. doi:10.1016/j.displa.2022.102240
- [56] Yiwen Wang, Ziming Li, Pratheep Kumar Chelladurai, Wendy Dannels, Tae Oh, and Roshan L Peiris. 2023. Haptic-Captioning: Using Audio-Haptic Interfaces to Enhance Speaker Indication in Real-Time Captions for Deaf and Hard-of-Hearing Viewers. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA,

- 1–14. doi:10.1145/3544548.3581076
- [57] Lauren A. Ward and Ben G. Shirley. 2019. Personalization in object-based audio for accessibility: A review of advancements for hearing impaired listeners. *Journal of the Audio Engineering Society* 67, 7/8 (Aug. 2019), 584. doi:10.17743/jaes.2019.0021
- [58] Carlee Wilson and Esther S Kim. 2021. Qualitative data collection: considerations for people with Aphasia. *Aphasiology* 35, 3 (March 2021), 314–333. doi:10.1080/02687038.2019.1693027
- [59] Jacob O. Wobbrock and Julie A. Kientz. 2016. Research contributions in human-computer interaction. *interactions* 23, 3 (April 2016), 38–44. doi:10.1145/2907069